

Fission in FISPACT-II

FISPACT-II workshop, NEA/OECD

Fission in FISPACT-II

- FISPACT-II is capable of including fission in inventory calculations but will not do so without additional inputs.
- If the **initial composition contains actinides fission should be included**. If a user is unsure, check the output file:

If **NUMBER OF FISSIONS** is greater than zero == a fission channel is open!

[tally of MT18 (n,f)]

→

NEUTRON FLUX DURING INTERVAL	1.00000E+13 n/cm ² /s	
NUMBER OF FISSIONS	7.71572E+14	BURN-UP OF ACTINIDES 2.31525E-08 %
INGESTION HAZARD FOR ALL MATERIALS	1.18118E+05 Sv/kg	
INHALATION HAZARD FOR ALL MATERIALS	2.65439E+05 Sv/kg	
INGESTION HAZARD EXCLUDING TRITIUM	1.18118E+05 Sv/kg	

To include fission:

- 1) Load Fission Yields (via the **files file/Nuclear Data Object**)
- 2) Turn on the fission process (**USEFISSION**)
- 3) Input which nuclides to be allowed/disallowed to undergo fission (**FISYEILD**)

Loading Fission Yields

Two types of fission yields in FISAPCT: Spontaneous and Induced.

- **Spontaneous yields** cover spontaneous fission. A decay process.
- **Induced fission yields** are those from (n,f) reactions (MT18)
- **NOFISS** keyword prevents yields from being read (*before FISPACT keyword*). *files file* Allows reuse of files file

FISPACT uses these to populate the inventory with the correct fission products

Loading via the *files* File:

```
# Library fission data
fy_endf /opt/fispact/nuclear_data/ENDFB71data/endfb71-n/endfb71nfy
sf_endf /opt/fispact/nuclear_data/ENDFB71data/endfb71-n/endfb71sfy
```

fy_endf : Induced fission yield files

sf_endf : Spontaneous fission yield file

Loading via the API:

```
ndr.setpath(pf.io.ND_FY_ENDF_KEY(), "/opt/fispact/nuclear_data/ENDFB71data/endfb71-n/endfb71nfy")
ndr.setpath(pf.io.ND_SF_ENDF_KEY(), "/opt/fispact/nuclear_data/ENDFB71data/endfb71-n/endfb71sfy")
```

Including these in a *files* file or API nuclear data object will cause FISAPCT-II to load the fission yield data.

Include fission in inventory

Fission can be turned on/off in FISPACT-II:

- In the input file:
 - **USEFISSION** keyword turn on fission (*after FISPACT keyword*).
- In the API, the **InputData** Object includes the **setusefission** method:
 - **True** for fission to be on, **False** to turn it off.
- By default, FISPACT-II uses independent (MT454) yields. To use the **Cumulative fission yield** data:
 - **CUMFYLD** keyword (*before FISPACT keyword*) should be included in the input file.
 - *setreadcumfy* to True in API NuclearDataReader object.

```
<< -----set initial conditions and get nuclear data----- >>
CLOBBER
MONITOR 1
GETXS 1 709
GETDECAY 1
CUMFYLD
FISPACT
* fission example
```

Example: use of CUMFLD in FISPACT-II input file

Example: use of setreadcumfy(True) in FISPACT Python API

```
ndr = pf.io.NuclearDataReader(m)
ndr.setpath(pf.io.ND_IND_NUC_KEY(), os.path.join(nd_base_path, 'ENDFB80data', 'endfb80_index'))
ndr.setpath(pf.io.ND_XS_ENDF_KEY(), os.path.join(nd_base_path, 'ENDFB80data', 'endfb80-n', 'gxs-709'))
ndr.setpath(pf.io.ND_FY_ENDF_KEY(), os.path.join(nd_base_path, 'ENDFB80data', 'endfb80-n', 'endfb80nfy'))
ndr.setpath(pf.io.ND_SF_ENDF_KEY(), os.path.join(nd_base_path, 'ENDFB80data', 'endfb80-n', 'endfb80sfy'))
ndr.setreadcumfy(True)
```

Including fission on Nuclides

- FISPACT-II will only allow fission to occur on a subset of nuclides.
- A user must specify which nuclides should be allowed to undergo fission via the **FISYIELD** keyword.
- Any number of nuclides can be set.

Input file syntax:

FISYIELD <number of nuclides>
nuclide 1
nuclide 2
etc.

If only 1 nuclide, can be single line: **FISYIELD 1 nuclide**

Example: *Include fission on U235 and U238 only*

```
<< -----set initial conditions and get nuclear data----- >>
CLOBBER
MONITOR 1
GETXS 1 709
GETDECAY 1
FISPACT
* fission example
<< -----add fission events to simulation----- >>
USEFISSION
FISYIELD 2
    U235
    U238
```

```
input.setusefission(True)
# list of nuclide zai's which fission should be included
input.setfissionyieldincludes( [ pf.util.zai_from_name(m, "U235"),
                                pf.util.zai_from_name(m, "U238") ] )
```

Example: *Include fission on U235 and U238 in Python API.*
*Here **input** is a **pf.InputData** object and **pf** is **pyfispact** module*

FISPACT API has **setfissionyieldincludes** method to set which takes a list of nuclides to use

Excluding fission on Nuclides

- Fission can be turned off for a subset of nuclides.
- If the **FISYIELD** keyword is invoked with a negative argument fission is disabled for the given nuclides.
- **FISYIELD 0** disables fission for all nuclides.
- Fission is allowed for all other nuclides.

```
<< -----add fission events to simulation----- >>
USEFISSION
FISYIELD -3
    Pu239
    Pu240
    Pu242
```

Example: Exclude fission on Pu239, Pu240 and Pu242.
Same input file syntax as before

- FISPACT API has *setfissionyieldexcludes* method for exclude fission on a set of nuclides.
- Also, *setfissionyieldexcludeall* method is available [equivalent of **FISYIELD 0**]

```
input.setusefission(True)
# list of nuclide zai's which fission should be excluded
input.setfissionyieldexcludes( [ pf.util.zai_from_name(m, "Pu239"),
                                pf.util.zai_from_name(m, "Pu240"),
                                pf.util.zai_from_name(m, "Pu242"), ] )
```

Example: Exclude fission on Pu239, Pu240 and Pu242
in FISPACT Python API.

All possible fissions

- All nuclides included in fission yields could also be added individually if desired. This will include all possible fissions
- However, the same result can be obtained via a little trick
- If fission is excluded on a fictitious nuclide, it will be included on all existing nuclides.
- Usually, we suggest that users exclude the non-existent nuclide **Es256m**:

```
<< -----add fission events to simulation----- >>  
FISYIELD -1 Es256m
```

Input file ↑

Example: Exclude fission on Es256m to include fission on all others.

```
input.setfissionyieldexcludes( [ pf.util.zai_from_name(m, "Es256m") ] )
```

Python API

Include fission in Pathways

Target nuclide I 131 98.188% of inventory given by 13 paths

path 1	48.925%	U 235	---	(R)---	Sb131	---	(d)---	Te131	---	(d)---	I 131	---	(S)---
			100.00%	(n,F)		100.00%	(b-)		100.00%	(b-)			
path 2	13.162%	U 235	---	(R)---	Sn131	---	(d)---	Sb131	---	(d)---	Te131	---	(d)---
			100.00%	(n,F)		100.00%	(b-)		100.00%	(b-)		100.00%	(b-)
path 3	13.162%	U 235	---	(R)---	Sn131m	---	(d)---	Sb131	---	(d)---	Te131	---	(d)---
			100.00%	(n,F)		100.00%	(b-)		100.00%	(b-)		100.00%	(b-)
path 4	5.473%	U 235	---	(R)---	Te131m	---	(d)---	I 131	---	(S)---			
			100.00%	(n,F)		100.00%	(b-)						
path 5	3.062%	U 235	---	(R)---	Te131	---	(d)---	I 131	---	(S)---			
			100.00%	(n,F)		100.00%	(b-)						
path 6	2.410%	U 235	---	(R)---	Sb131	---	(d)---	Te131m	---	(d)---	I 131	---	(S)---
			100.00%	(n,F)		100.00%	(b-)		100.00%	(b-)			
path 7	1.913%	U 235	---	(R)---	Te131m	---	(b)---	Te131	---	(d)---	I 131	---	(S)---
			100.00%	(n,F)		100.00%	(IT)		100.00%	(b-)			
							0.00%	(n,E)					
							0.00%	(n,n)					
path 8	0.842%	U 235	---	(R)---	Sb131	---	(d)---	Te131m	---	(b)---	Te131	---	(d)---
			100.00%	(n,F)		100.00%	(b-)		100.00%	(IT)		100.00%	(b-)
									0.00%	(n,E)			
									0.00%	(n,n)			
path 9	0.648%	U 235	---	(R)---	Sn131m	---	(d)---	Sb131	---	(d)---	Te131m	---	(d)---
			100.00%	(n,F)		100.00%	(b-)		100.00%	(b-)		100.00%	(b-)
path 10	0.648%	U 235	---	(R)---	Sn131	---	(d)---	Sb131	---	(d)---	Te131m	---	(d)---
			100.00%	(n,F)		100.00%	(b-)		100.00%	(b-)		100.00%	(b-)
path 11	2.743%	U 238	---	(R)---	Sn131m	---	(d)---	Sb131	---	(d)---	Te131	---	(d)---
			100.00%	(n,F)		100.00%	(b-)		100.00%	(b-)		100.00%	(b-)
path 12	2.743%	U 238	---	(R)---	Sn131	---	(d)---	Sb131	---	(d)---	Te131	---	(d)---
			100.00%	(n,F)		100.00%	(b-)		100.00%	(b-)		100.00%	(b-)
path 13	2.457%	U 238	---	(R)---	Sb131	---	(d)---	Te131	---	(d)---	I 131	---	(S)---
			100.00%	(n,F)		100.00%	(b-)		100.00%	(b-)			

- Fission can be included in FISPACT-II's uncertainty-pathway analysis
➤ *[see presentation on Uncertainties and Pathways]*
- This is done with the **FISCHOOSE** keyword. Once again which nuclides should be included is given by the user.

```
FISCHOOSE 2
U235
U238
UNCERTAINTY 2
```

- Allows only specific fission pathways to be included.
- Uncertainties and Pathways are currently not available in the FISPACT API.

Example: Pathways to I131 from U235 and U238

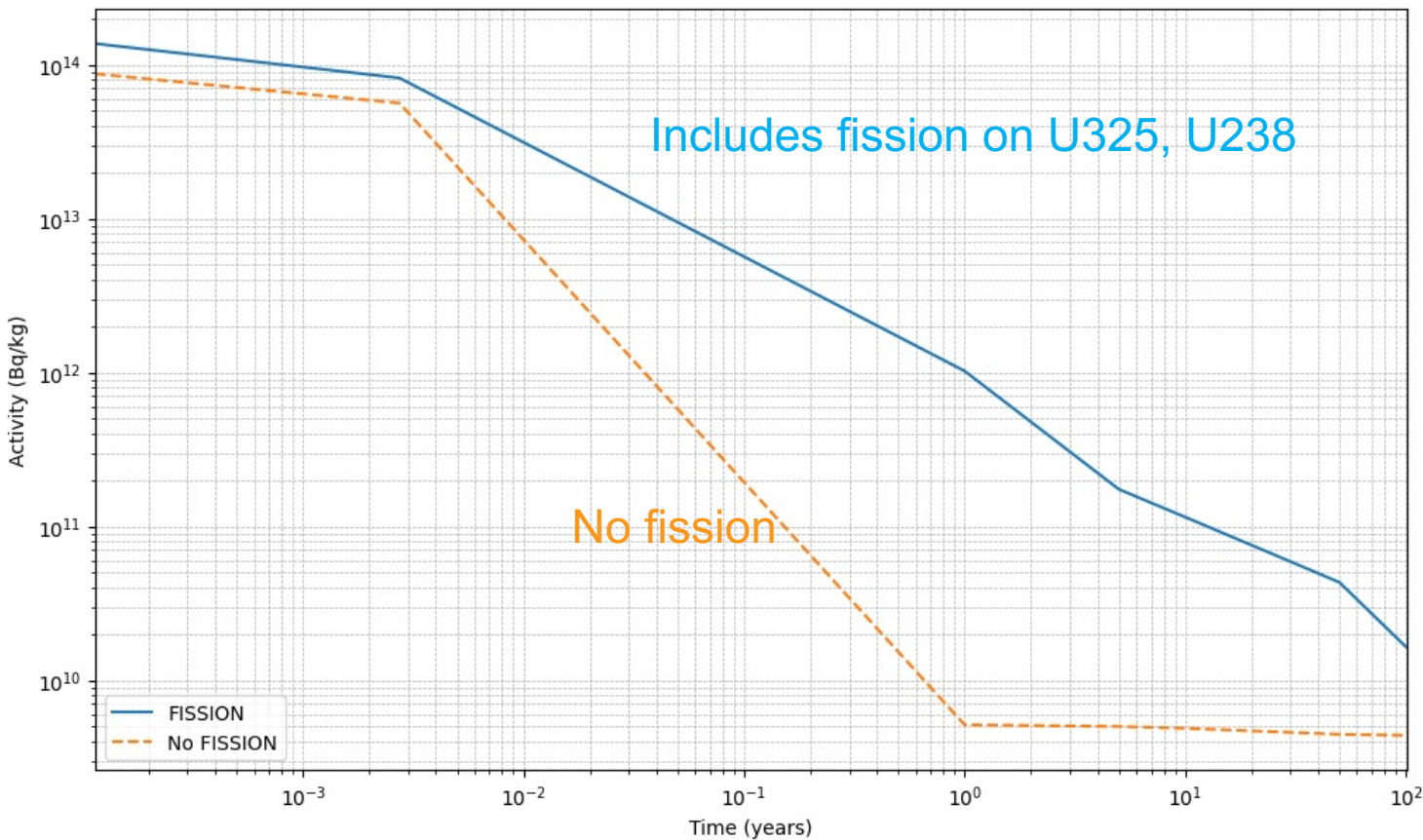
Example

Irradiation of UO_2 (3% $\text{U}235$) fuel with a PWR spectrum:

- Only allow fission on $\text{U}235$ and $\text{U}238$ for this example.
- Spectrum taken from FISPACT reference spectra:
https://fispact.ukaea.uk/wiki/Reference_input_spectra
- Use Nuclear Data from ENDF/BVIII library (xs, decay and fission yields).
 - Use independent induced fission yields
- Irradiate material for 1 year.
- Output inventory every 3 months during irradiation.
- Let inventory cool for 100 years:
 - Cooling times: 1hr, 1day, 1year, 5years, 10years, 50years, 100years

Results

Including fission in an inventory calculation will produce the correct fission products and therefore correct inventory quantities



Time after irradiation

ATOMS		
1	H	2.3913E+16
2	He	2.1716E+19
3	Li	3.5297E+06
4	Be	2.2458E+12
5	B	1.0614E+08
6	C	7.9894E+18
7	N	2.2959E+17
8	O	2.0323E+25
9	F	3.1863E+17
10	Ne	6.4487E+10
28	Ni	5.6487E+06
29	Cu	3.5312E+10
30	Zn	2.3650E+14
31	Ga	3.2761E+14
32	Ge	5.1760E+16
33	As	1.5253E+16
34	Se	7.1658E+18
35	Br	2.7170E+18
36	Kr	4.1638E+19
37	Rb	4.8689E+19
38	Sr	5.4280E+19
39	Y	6.3817E+19
40	Zr	4.9483E+20
41	Nb	3.9579E+15
42	Mo	3.4062E+20
43	Tc	8.5157E+19
44	Ru	1.6460E+20
45	Rh	4.5379E+19
46	Pd	3.0073E+19
47	Ag	6.9413E+17
48	Cd	1.3755E+18
49	In	2.3301E+17
50	Sn	2.5474E+18
51	Sb	5.3989E+17
52	Te	3.0930E+19
53	I	1.0438E+19
54	Xe	3.0712E+20
55	Cs	1.8599E+20
56	Ba	1.7010E+20
57	La	8.8449E+19
58	Ce	1.6602E+20
59	Pr	8.0835E+19
60	Nd	2.8658E+20
61	Pm	7.1251E+08
62	Sm	5.5492E+19
63	Eu	5.5664E+18
64	Gd	1.0583E+18
65	Tb	2.3195E+16
66	Dy	4.3065E+15
67	Ho	1.4558E+14
68	Er	2.0623E+14
69	Tm	2.3278E+13
70	Yb	1.8404E+13
81	Tl	1.9159E+06
82	Pb	6.8693E+12
83	Bi	4.5390E+08
84	Po	1.7233E+09
86	Rn	8.6007E+07
87	Fr	1.2248E+05
88	Ra	1.3156E+13
89	Ac	4.6195E+12
90	Th	3.1557E+16
91	Pa	9.9060E+15
92	U	3.3267E+24
93	Np	6.7185E+18
94	Pu	4.4770E+21
95	Am	8.7786E+17
96	Cm	1.7528E+10

Inventory atoms by element with fission

Taken from COMPOSITION OF MATERIAL BY ELEMENT section of FISPACT-II output file

Inventory atoms by element without fission

ATOMS		
1	H	6.0873E+15
2	He	2.0584E+18
3	Li	2.2171E+05
4	Be	4.8838E+11
5	B	4.5288E+06
6	C	1.9977E+18
7	N	5.7048E+16
8	O	2.0323E+25
9	F	7.9656E+16
10	Ne	4.0304E+09
82	Pb	8.0456E+05
88	Ra	1.3510E+08
89	Ac	6.4664E+07
90	Th	3.3960E+14
91	Pa	2.5119E+13
92	U	3.3314E+24
93	Np	4.3211E+19
94	Pu	1.0813E+21
95	Am	4.5811E+13
96	Cm	6.0258E+10

Summary and extension

Fission can be included in FISPACT-II simulations by loading the required data and using the related keywords.

- Include fission yield sin files file or API nuclear data object
- Include **USEFISSION** keyword or *Inputdata.setusefission* to **True**
- Define which nuclides should under fission:
 - **FISYIELD** < number of nuclides > <list of nuclides >.
 - < number of nuclides > Positive for include, Negative for exclude
 - API use *Inputdata.setfissionyieldincludes* and *Inputdata.setfissionyieldexclude*

Extensions to example:

- What if all possible fissions were included?
- How does using Cumulative yields effect the results?